

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Observation Task No.	7
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1. TITLE

NPM and package.json in Node.js

2. AIM

To understand NPM, package.json, dependencies and the process of installing and using an external Node.js package.

3. OBJECTIVES

1. Understand the role of NPM.
2. Understand package.json and dependency management.
3. Initialize a Node.js project.
4. Install an external package.
5. Use the installed package in a Node.js application.

4. THEORY

NPM is the package manager commonly used with Node.js. The package.json file stores project metadata, scripts and dependencies. Installing an external package adds it to the project's dependency information and makes it available to the application through `require()` or `import`.

5. ALGORITHM / PROCEDURE

1. Create a project folder and open it in VS Code.
2. Run `npm init -y` to create package.json.
3. Install the external validator package using `npm install validator`.
4. Create `app.js` and require the package.
5. Use `validator.isEmail()` to validate sample email addresses.
6. Run the application using `node app.js` or `npm start`.

6. PROGRAM

Source file: app.js

```
// Task 7: NPM and package.json

// validator is an external package installed through NPM.
const validator = require("validator");

const email = "student@example.com";
const invalidEmail = "student@example";

console.log("NPM AND PACKAGE.JSON DEMONSTRATION");
console.log("Package used: validator");

console.log("\nValid email:", email);
console.log("Validation result:", validator.isEmail(email));

console.log("\nInvalid email:", invalidEmail);
console.log("Validation result:", validator.isEmail(invalidEmail));
```

7. CODE / SYNTAX EXPLANATION

Code / Syntax	Explanation
npm init -y	Creates package.json with default project metadata.
npm install validator	Downloads the external package and records it as a dependency.
package.json	Stores project metadata, scripts and dependencies.
node_modules	Contains installed package files.
require('validator')	Loads the installed package into the program.
validator.isEmail()	Checks whether a string has a valid email format.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Status
TC-01	npm init -y	package.json created	Pass
TC-02	npm install validator	Dependency installed	Pass
TC-03	student@example.com	Validation returns true	Pass
TC-04	student@example	Validation returns false	Pass

9. OUTPUT

Sample execution output. Install the required external package and run the program locally before final submission.

Task 7 - Execution Output

```
NPM AND PACKAGE.JSON DEMONSTRATION
Package used: validator

Valid email: student@example.com
Validation result: true

Invalid email: student@example
Validation result: false
```

Output summary:

NPM AND PACKAGE.JSON DEMONSTRATION
Package used: validator

Valid email: student@example.com
Validation result: true

Invalid email: student@example
Validation result: false

10. RESULT

Thus, NPM and package.json were demonstrated by installing an external package and using its validation method in a Node.js application. The sample output in this observation is the expected application output; run npm install validator locally before final submission.